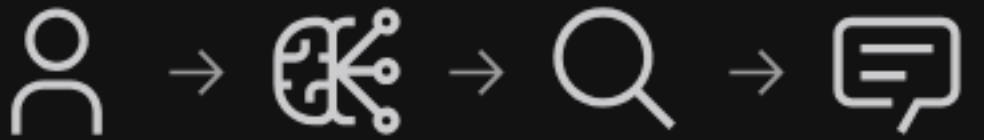
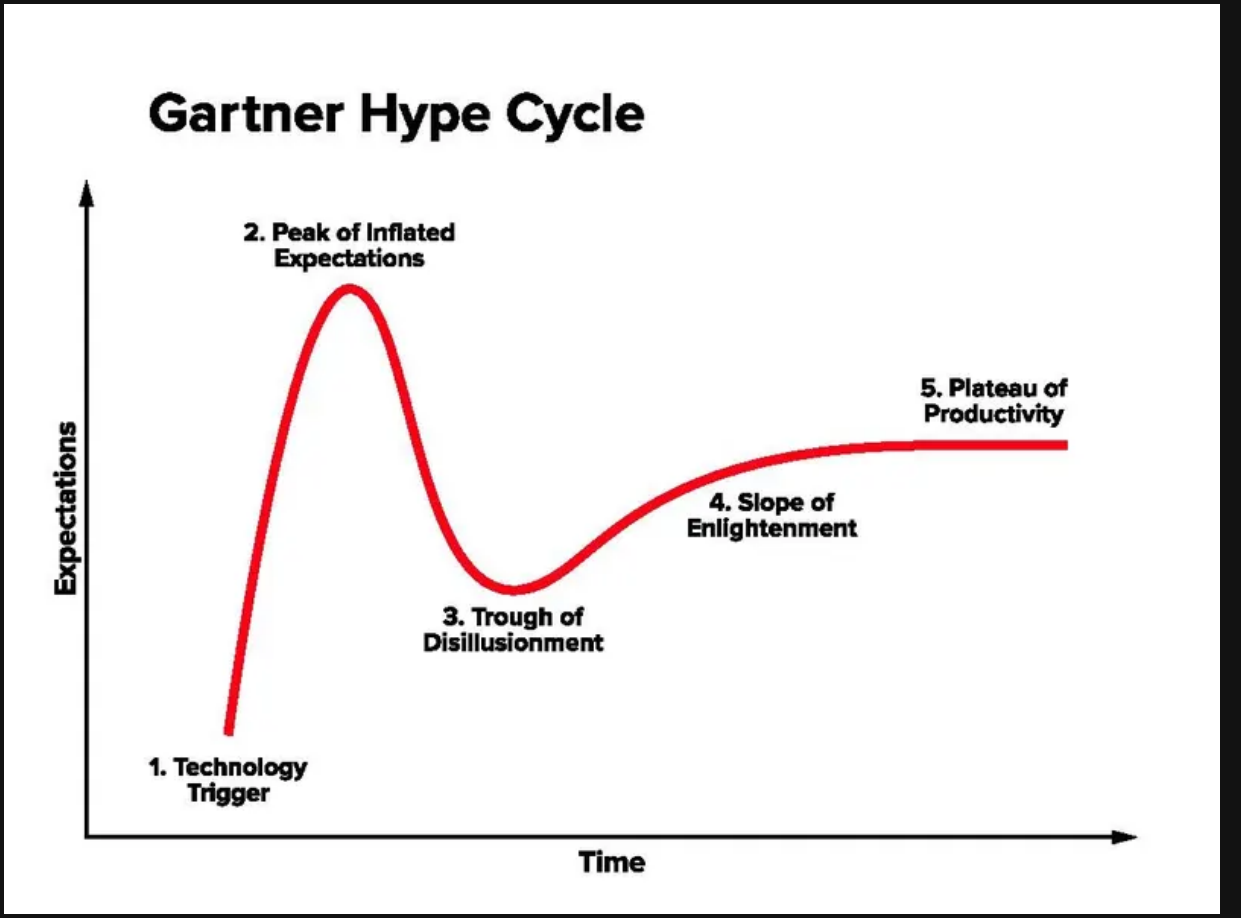
LLM Workflows



Prepared for Wheaton College: COMP 255

Ben Batorsky

AI coding assistance was used in developing these materials



John Werner, "The Trough Of Disillusionment And Four Outliers On The Gartner Hype Cycle", *Forbes*, 18 Jul. 2024.





Large Language Model (LLM)

Large Language Model (LLM)



Text Corpus

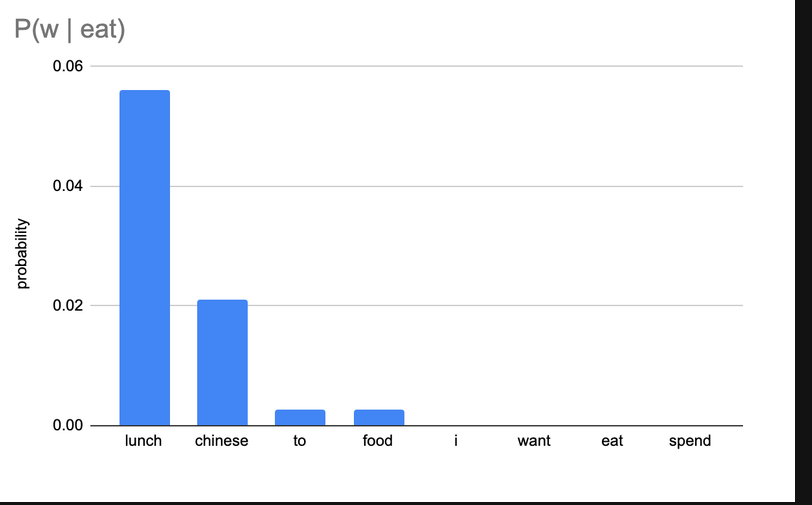


Counter

	i	want	to	eat	chinese	food	lunch	spend
i	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Figure 3.1 Bigram counts for eight of the words (out of $V = 1446$) in the Berkeley Restaurant Project corpus of 9332 sentences. Zero counts are in gray. Each cell shows the count of the column label word following the row label word. Thus the cell in row **i** and column **want** means that **want** followed **i** 827 times in the corpus.

I want to eat ___


P(w | eat)
0.06
0.04
0.02
0.00
probability
lunch
chinese
to
food
i
want
eat
spend


Corpus
I want to eat ____
Neural Network (NN)
Predict next word
lunch
spend
Trained NN

Large Language Model (LLM)

A model that predicts the next word given context.

Large Language Model (LLM)

Language Model



Corpus Trained LM

Language Model

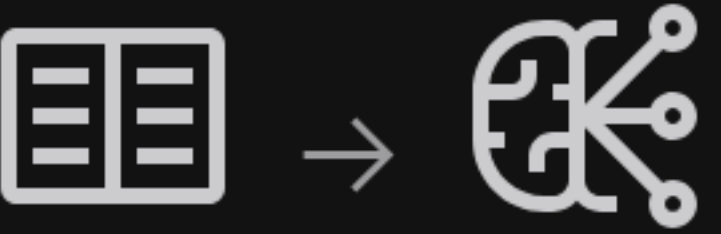


Corpus **Trained LM**

Text: Billions

Parameters: Millions

LLM

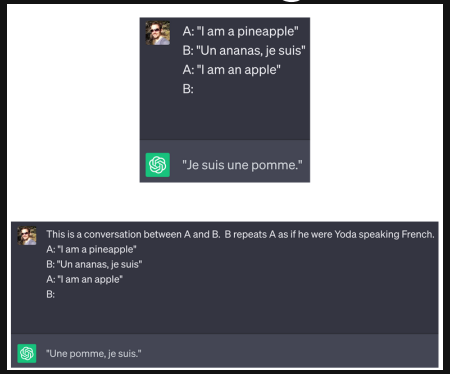


Corpus **Trained LM**

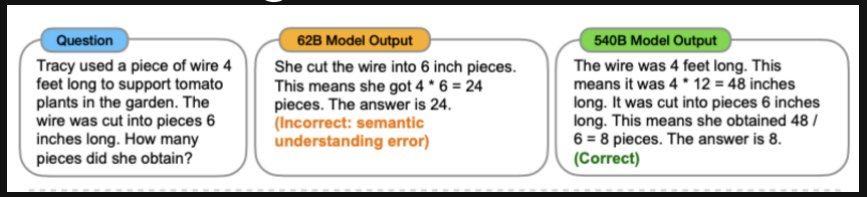
Text: Trillions

Parameters: Trillions

In-context learning



Reasoning



Characterizing Emergent Phenomena in Large Language Models – Google AI Blog

Notebook: Pre-trained models

I am a data scientist and I **am always looking to improve the data processing abilities of people who are passionate about data science.**

Written by Transformer · transformer.huggingface.co 🦄

More info on alignment tuning via Reinforcement Learning from Human Feedback (see appendix)

Notebook: Post-trained models

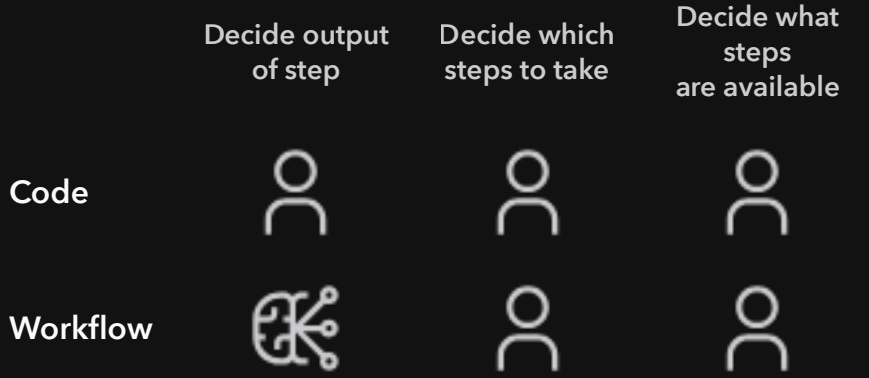
Large Language Models

Typically refers to models that generate useful responses to inputs.

Anthropic: Systems where LLMs and tools are orchestrated through predefined code paths.

Requires:

- "State"
- Tools



Adapted from Building effective agents, Anthropic; table from What is an AI agent?

Notebook: State as conversation history

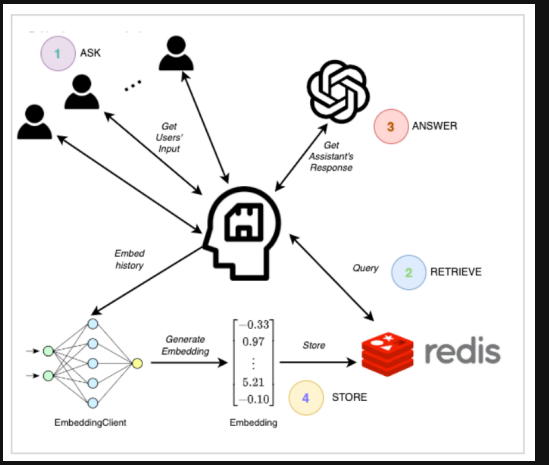
You
What day is it today

ChatGPT
Today is February 6th, 2024.



You
What day was it the last time we spoke?

ChatGPT
The last time we spoke was on February 6th, 2024, which was a Tuesday.



Introducing the ChatGPT Memory Project – Redis

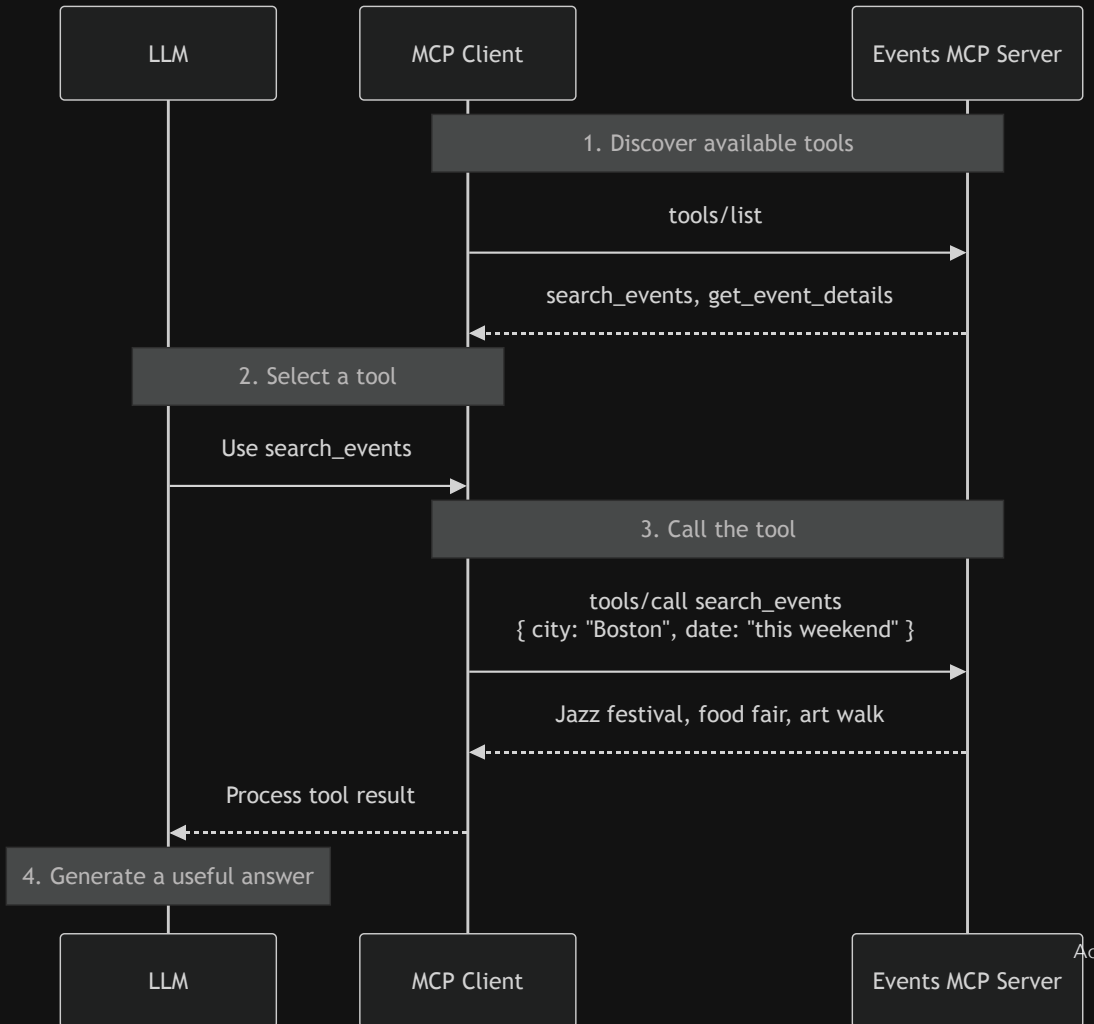

abc
Text
LLM
?



So how does an LLM use tools?

Notebook: Tools

User: “What events are happening in Boston this weekend?”



Adapted from Model Context Protocol – Tools



Patient

"I'm feeling not good today, I am coughing and sneezing"

Parse symptoms

Web search

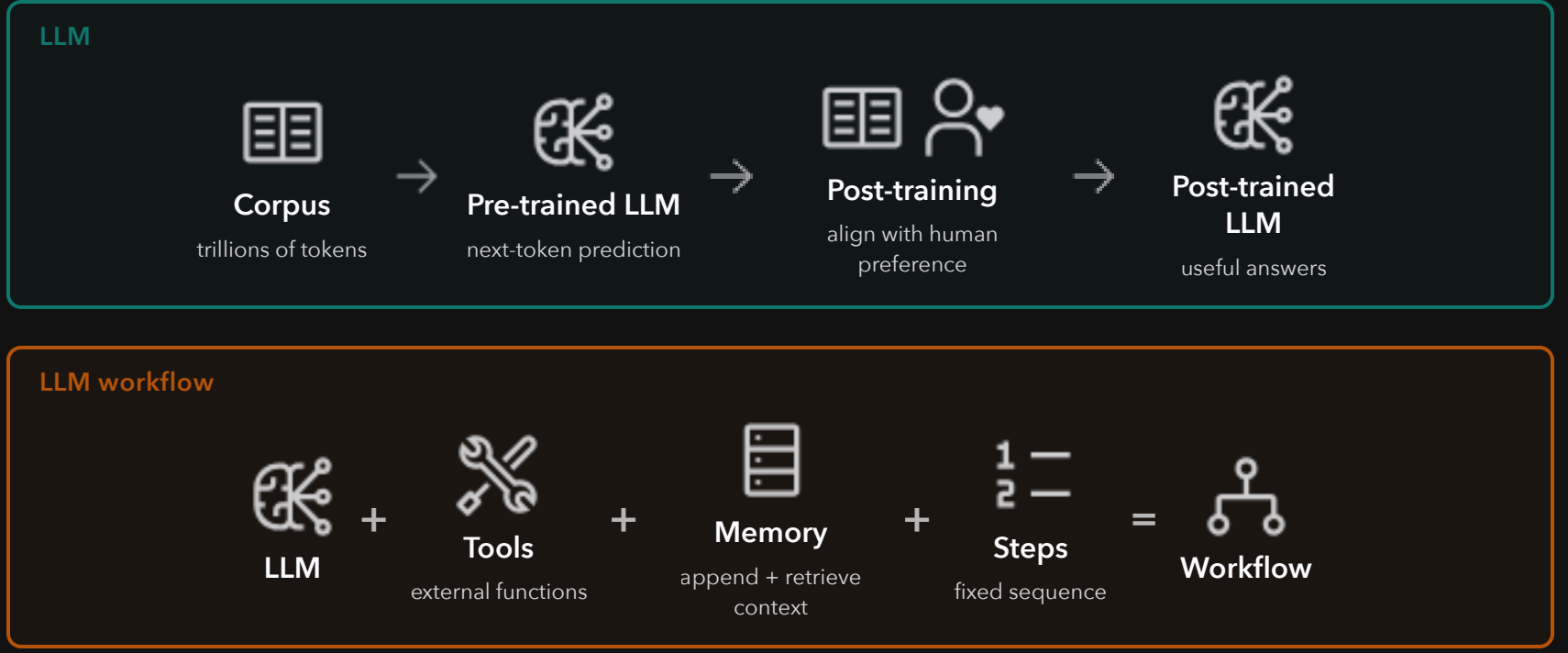
tool call

Diagnose

Save record

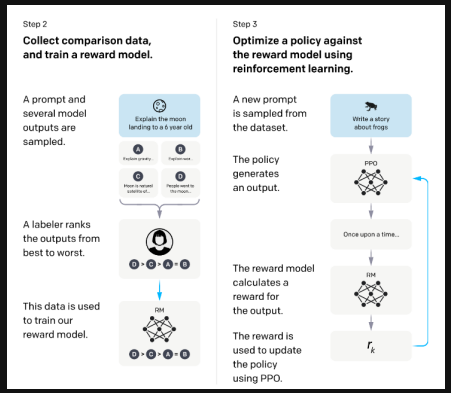
Respond

Notebook: Healthbot Workflow


LLM
Corpus
trillions of tokens
Pre-trained LLM
next-token prediction
Post-training
align with human preference
Post-trained LLM
useful answers
LLM workflow
LLM
+
Tools
external functions
+
Memory
append + retrieve context
+
Steps
fixed sequence
=
Workflow

Appendix

(RLHF, 2022)



[2203.02155] Training language models to follow instructions with human feedback